Connect the forces to the final position

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Constant net force / ES: fuerza neta constante / PT: força resultante constante

Gives constant acceleration: a straight v–t line.

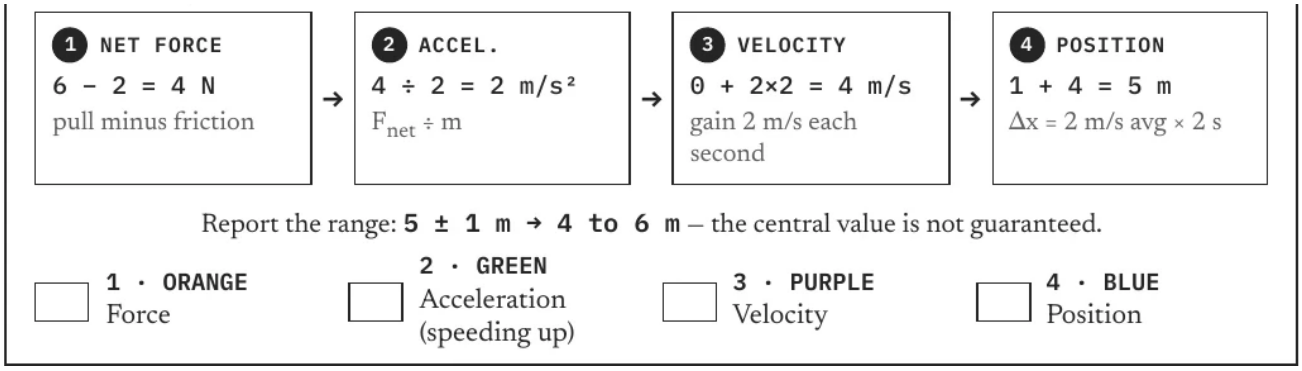
Also on your note page: Force and motion, Zero net force, Range.

B READ THE RELATIONSHIP

Net force determines acceleration. To find position, you also need the starting motion.

Force → acceleration → velocity → position

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart has zero net force and already moves at +2 m/s. After 3 s, find its velocity and displacement.

Given F net = 0 N v = +2 m/s t = 3 s

Rule a = 0 → v constant $\Delta x = v \times t$

Work v stays +2 m/s $\Delta x = 2 \times 3$

Answer +2 m/s, +6 m

E SAVE FOR THE FINAL MEETING

The chain is complete: net force → acceleration → velocity → position. That is the whole route from the data on XJ to a forecast of where it will arrive.

Read the physics: Constant net force

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Divide net force by mass to find acceleration. For constant acceleration, use the initial velocity and elapsed time to find final velocity.

02 HOW IT WORKS

Use the signed area under the velocity–time graph for displacement. Add displacement to starting position. A section below zero subtracts from displacement but still counts toward distance.

03 WHAT TO CHECK

With zero net force a moving object continues at its initial velocity. With constant nonzero net force its velocity changes by the same amount each second.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart has zero net force and already moves at +2 m/s. After 3 s, find its velocity and displacement. +2 m/s, +6 m.

05 CONNECT TO XJ

The chain is complete: net force → acceleration → velocity → position. That is the whole route from the data on XJ to a forecast of where it will arrive.

SAY IT BEFORE YOU WRITE

What do you need besides force and mass to predict position? From force I can find ___. To find position I also need ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Force and motion

ES: fuerza y movimiento / PT: força e movimento

Meaning in my words	How can you tell?
How does a net force change motion? In my words, this means ___.	A cart feels a net force right. What else tells you whether it is speeding up right now? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch a cart moving left while a net force points right. Which way does acceleration point? My example shows ___.	"The force points right, so the cart is already going right." What evidence is missing? I would change ___ because ___.

Constant net force

ES: fuerza neta constante / PT: força resultante constante

Meaning in my words	How can you tell?
How would you explain constant net force to a friend? In my words, this means ___.	If a cart's mass and its net force both stay fixed, what stays fixed about the motion? I can tell because ___.
Example or sketch	**Non-example and why**
A 2 kg cart feels 4 N right throughout two seconds. What acceleration would you label? My example shows ___.	Does constant net force always mean steady velocity? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Zero net force

ES: fuerza neta nula / PT: força resultante nula

Meaning in my words	How can you tell?
How would you explain zero net force to a friend? In my words, this means ___.	What happens to a moving cart's velocity if its forces add to zero? I can tell because ___.
Example or sketch	**Non-example and why**
Draw a cart moving right at 2 m/s a second after its horizontal forces balance and stay balanced. My example shows ___.	Someone leaves every force off a moving cart because its net force is zero. What could still act? I would change ___ because ___.

Range

ES: intervalo / PT: intervalo

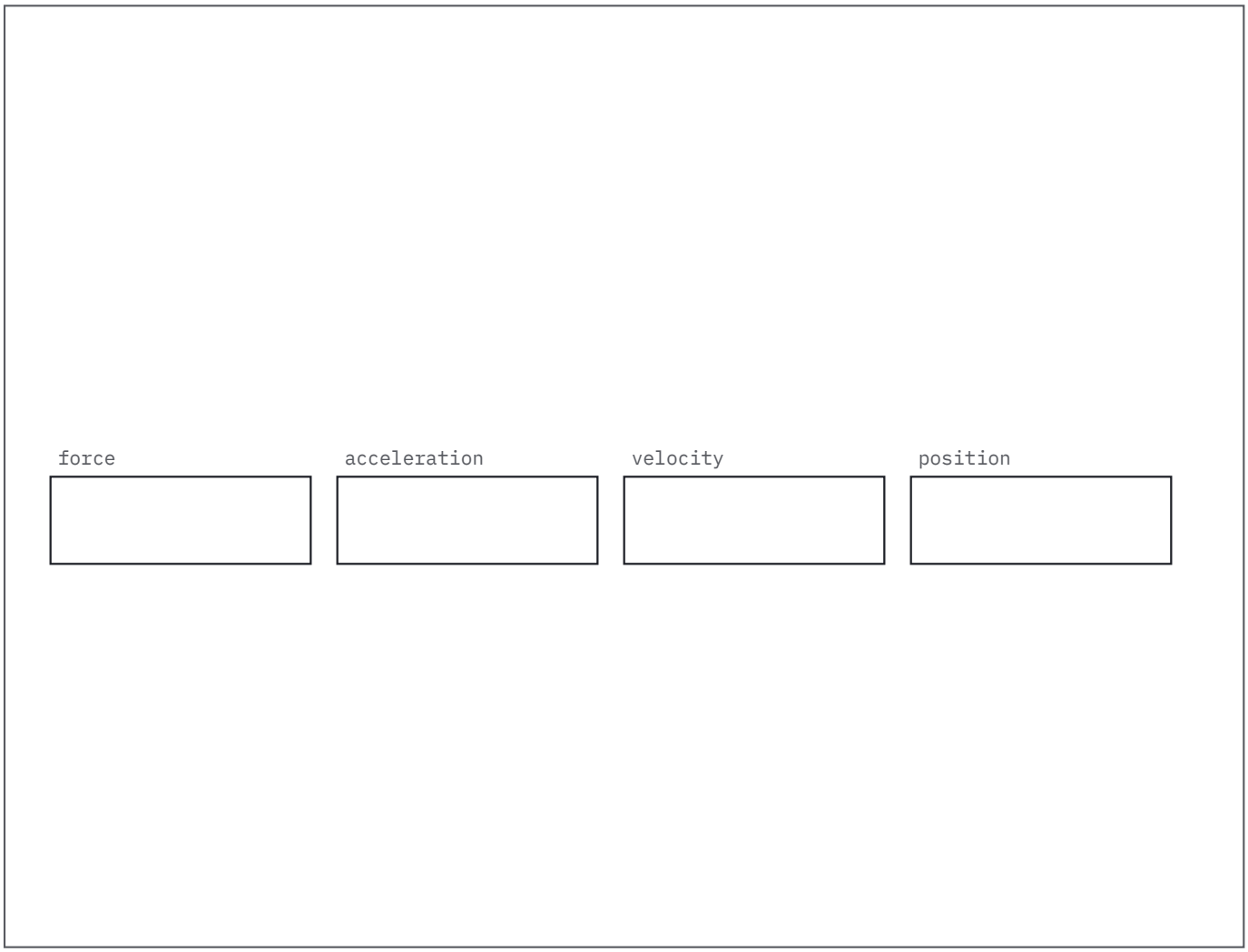
Meaning in my words	How can you tell?
How would you explain range to a friend? In my words, this means ___.	What positions are included in 12 ± 2 m? I can tell because ___.
Example or sketch	**Non-example and why**
How would you mark the forecast on a number line? My example shows ___.	A partner says only 12 m counts as a match. What do the endpoints show? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Build a four-frame chain from force to position. Add the missing inputs and units beside the arrows.

Labels you can use: force · velocity · position.



Tell a partner

From force I can find ___. To find position I also need ___.

Can we release a forecast?

20

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Identify which measurements a position forecast needs.

CHOOSE ONE JOB TOGETHER

MISSION SYSTEMS ENGINEER ☐

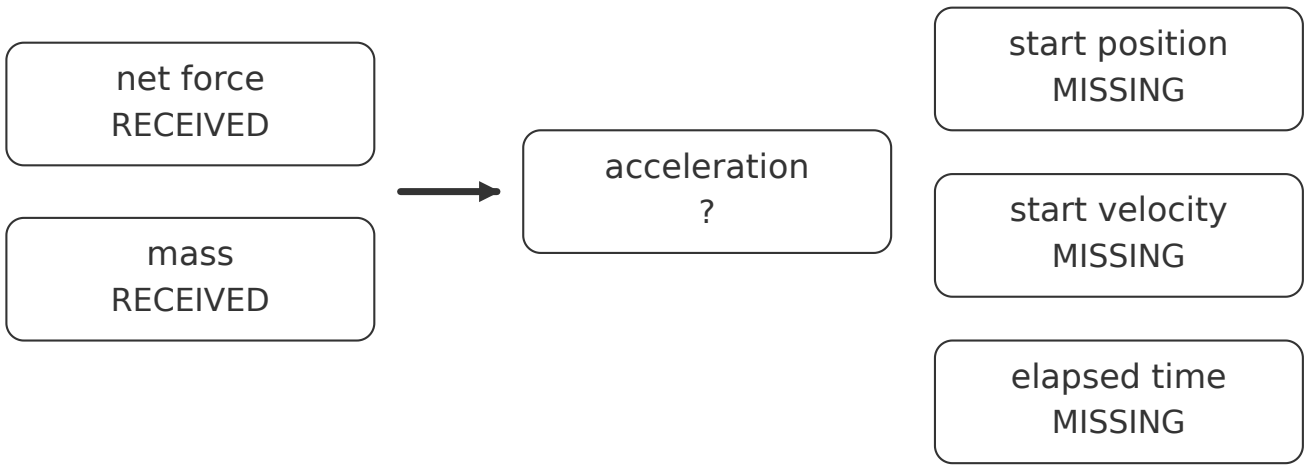
Mark what you can calculate now. Send the telescope operator a request for the measurements still needed.

NEWS ASSIGNMENT EDITOR ☐

Prepare a 'known / still needed' graphic. Tell the anchor which result can air and which must wait.

Word help: known / missing / forecast

A telescope manager wants a final position. The telemetry feed contains net force and mass, but no starting velocity, starting position or elapsed time.



01 OUR FIRST CALL

What can the desk release now? What measurements must the crew still collect?

We can find ___. We still need ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Is the forecast ready after the update? Say what you can release and what is still missing.

Now we would say ___ because ___.

Switch speakers. What can we calculate, and what is still missing?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks